From Marchenko–Pastur to Woodbury: Direct Solvers for Long-Only Mean-Variance Portfolios

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Abstract

The sample covariance matrix of equity returns is severely ill-conditioned: a small number of pervasive risk factors dominate the eigenvalue spectrum while several hundred noise eigenvalues crowd near zero. Random matrix theory identifies the signal factors via the Marchenko–Pastur bulk edge and replaces the noise eigenvalues with their average, yielding the trace-preserving "constant residual eigenvalue" (CRE) estimator $T_0^{\mathrm{RMT}} = \bar{\lambda} I + U_k \Delta_k U_k^\top$, which on a five-year S&P 500 sample improves the condition number by a factor of roughly 300.

We exploit the low-rank-plus-identity structure of this estimator: the Woodbury identity gives the exact inverse of the active-set system at $O(n_a k^2 + k^3)$ per outer step of a primal-dual active-set method, replacing the $O(n_a^2 k + n_a^3)$ assemble-and-factor approach. The Woodbury solve requires fewer flops precisely when $k < n_a$, a condition that holds throughout the long-only equity regime. The requisite eigenpairs are computed once by a randomised SVD applied directly to the return matrix at $O(nTk)$, without forming $\hat{\Sigma}$, so that for a sweep of S related solves the preprocessing amortises to $O(nTk/S)$ per solve.

On efficient-frontier sweeps the measured advantage over the assemble-and-factor baseline is fivefold warm and roughly twentyfold cold, growing with n; the advantage over a tuned interior-point baseline exceeds two orders of magnitude, and an exact critical-line sweep built on the same kernel traces every frontier breakpoint for the cost of a coarse grid. The returned weights match interior-point solutions to solver accuracy, the Marchenko–Pastur threshold is shown to be an insensitive hyperparameter, and rolling out-of-sample backtests on two universes confirm that the cleaned estimator carries no measurable statistical penalty relative to standard shrinkage.

1 Introduction

The minimum-variance portfolio is the solution to a quadratic programme whose inputs are a covariance estimate and a set of weight constraints. In the large-n regime of institutional equity portfolios ($n/T \approx 0.4$ for a five-year daily window), the sample covariance $\hat{\Sigma} = X^\top X/T$ is nearly singular, with condition number $\kappa(\hat{\Sigma}) \approx 106{,}000$ on S&P 500 data. This ill-conditioning makes the portfolio optimisation problem numerically fragile.

Random matrix theory (RMT) attributes most of this ill-conditioning to noise: the Marchenko–Pastur (MP) law identifies eigenvalues below the bulk edge $\hat{\lambda}_+$ as uninformative, consistent with noise, while eigenvalues above $\hat{\lambda}_+$ carry genuine factor structure. The "constant residual eigenvalue"

(CRE) cleaning method [9, 3, 2] replaces the noise eigenvalues with their average $\bar{\lambda}$ – preserving the trace of $\hat{\Sigma}$ – and produces the estimator

$$T_0^{\mathrm{RMT}} = \bar{\lambda} I + U_k(\Lambda_k - \bar{\lambda} I)U_k^\top,$$

where (U_k, Λ_k) are the k eigenpairs above $\hat{\lambda}_+$. On S&P 500 data, $k = 23$ and $\kappa(T_0^{\mathrm{RMT}}) \approx 343$ – roughly 300 times better conditioned than $\hat{\Sigma}$.

Implementations of CRE cleaning in the public domain typically form T_0^{RMT} as a dense $n \times n$ matrix and pass it to a general-purpose solver (CVXPY, interior-point, etc.); see e.g. the reference implementations accompanying [14]. This paper exploits the low-rank-plus-identity structure $T_0^{\mathrm{RMT}} = \bar{\lambda} I + U_k \Delta_k U_k^\top$, which renders the matrix inverse available in closed form via the Woodbury identity, without assembling either the $n \times n$ covariance or the $n_a \times n_a$ active-set restriction. The result is a direct active-set portfolio solver that is substantially faster than both general-purpose interior-point solvers and the KKT-Cholesky approach of assembling the restricted system and factoring. Exploiting factor structure in portfolio quadratic programmes via the Sherman–Morrison–Woodbury identity goes back at least to Perold [19]; the contribution here is the combination with RMT-determined structure, randomised preprocessing, and an end-to-end complexity and accuracy account.

Contributions This paper makes four contributions, each self-contained relative to the companion paper [21]. The companion paper uses a different estimator structure and a different inner solver; the Woodbury treatment, the crossover analysis, the rSVD preprocessing treatment, and the total pipeline complexity accounting developed here are absent from it.

First, we give a complete treatment of the randomised SVD preprocessing step (Section 5). The companion paper uses a dense eigendecomposition of $\hat{\Sigma}$ for simplicity; at large n this costs $O(n^2T + n^3)$ and re-introduces the n^2 storage the matrix-free approach aims to avoid. A randomised truncated SVD applied directly to X computes the top-k eigenpairs of $\hat{\Sigma}$ in $O(nTk)$ time and $O(nk)$ storage; the measured portfolio impact of the approximation is below three basis points in any weight. For daily rolling windows we additionally implement an incremental rank-two update of the tracked eigenpairs that avoids re-sketching altogether.

Second, we derive the exact crossover condition for Woodbury vs. KKT-Cholesky (Proposition 4.2): the Woodbury inner solve requires fewer flops per outer step precisely when $k < n_a$, a condition satisfied throughout the long-only equity regime (k/n_a between roughly 0.2 and 0.7 in our experiments). We also account for the total pipeline cost over a sweep of S related solves (Proposition 4.3), under which the $O(nTk)$ preprocessing amortises to $O(nTk/S)$ per solve.

Third, we provide a direct numerical comparison of Woodbury against KKT-Cholesky, a tuned interior-point baseline (Clarabel called natively on pre-assembled data), and CVXPY on the same RMT problem (Section 7), together with optimality cross-checks of the returned weights, an exact critical-line sweep built on the same Woodbury kernel that traces every frontier breakpoint (Section 7.4), a k-sensitivity audit showing that the MP threshold choice is not a sensitive hyperparameter (Section 6), and an out-of-sample backtest on two universes – S&P 500 and FTSE 100 – showing that the cleaned estimator carries no measurable statistical penalty relative to standard shrinkage (Section 7.5).

Fourth, we delineate the scope of the structural trick (Remarks 3.2 and 4.2): ridge terms, diagonal loadings, externally specified factor models, and linear equality constraints preserve the Woodbury structure, whereas statistically richer cleanings – nonlinear shrinkage and general rotationally invariant estimators – do not.

Relation to existing work Laloux et al. [9] and Plerou et al. [20] introduced the Marchenko–Pastur threshold to financial covariance estimation; Bouchaud and Potters [2] develop the theory in detail. Bun, Bouchaud and Potters [3] give a comprehensive treatment of CRE cleaning and of the statistically superior rotationally invariant estimators built on the eigenvector overlaps of Ledoit and Péché [12]; nonlinear shrinkage is developed by Ledoit and Wolf [13, 11]. Those estimators modify the full spectrum and are not Woodbury-compatible; this paper works with CRE precisely because clipping is the member of the family with exploitable algebraic structure (Remark 3.2). López de Prado [14] provides accessible CRE implementations for practitioners, built on dense covariance assembly and general-purpose solvers.

On the optimisation side, Perold [19] exploited diagonal-plus-low-rank factor structure inside a large-scale portfolio QP solver four decades ago, and the critical line algorithm of Markowitz [16] (see [17, 22] for modern treatments) traces the exact long-only frontier by active-set pivoting. Relative to this literature our contribution is the pairing of the *RMT-determined* low-rank structure with the active-set solve, the randomised-SVD preprocessing with an explicit total-pipeline complexity accounting, and the measured end-to-end comparison (Remark 4.1 discusses the relation to CLA). The randomised SVD itself is due to Halko, Martinsson and Tropp [6] and has been applied widely to PCA and large-scale covariance estimation.

This paper is self-contained: Algorithm 1 gives the complete primal-dual active-set solver including the Woodbury inner solve, and Section 5 develops the rSVD preprocessing independently. A companion paper [21] addresses the same portfolio problem with an iterative inner solver and a different covariance structure.

Outline Section 2 states the problem. Section 3 constructs the cleaned estimator. Section 4 develops the Woodbury portfolio algorithm and the crossover analysis. Section 5 covers randomised SVD preprocessing. Section 6 analyses sensitivity to the threshold k. Section 7 reports timing, scaling, and out-of-sample results. Section 8 concludes.

2 Problem Setup

Let $X \in \mathbb{R}^{T\times n}$ be the demeaned return matrix for n assets over T periods. The long-only mean-variance portfolio solves

$$\min_{w}\; w^\top T_0^{\mathrm{RMT}} w - \rho\,\hat{\mu}^\top w \quad \text{subject to} \quad \mathbf{1}^\top w = 1, \quad w \geq 0\,,$$

where $\rho \geq 0$ is a return-tilt weight, $\hat{\mu}$ is a vector of expected returns, and T_0^{RMT} is the RMT-cleaned covariance estimator defined in Section 3.[1]

Remark 2.1 (Why not blend with the sample covariance?)**.** *A general shrinkage estimator $\hat{\Sigma}_\alpha = (1-\alpha)\hat{\Sigma} + \alpha T_0^{\mathrm{RMT}}$ with $\alpha \in (0,1)$ would re-introduce a fraction of the bulk eigenvalues that the cleaning replaces. Our reason for fixing $\alpha = 1$ is algebraic, not statistical: only at $\alpha = 1$ does the system matrix retain the low-rank-plus-identity form that the Woodbury solve of Section 4 requires. For $\alpha \in (0,1)$ the matrix $(1-\alpha)\hat{\Sigma}_{\mathcal{A}} + \alpha T_{0,\mathcal{A}}^{\mathrm{RMT}}$ has full-rank residual structure – no $k \times k$ correction matrix exists – and the $O(n_a k^2)$ cost advantage is entirely lost. The statistical case for CRE cleaning itself rests on the RMT literature (Section 3); Section 7.5 reports out-of-sample evidence that the cleaned estimator is competitive with standard shrinkage on the data used in this paper.*

[1] The symbol T_0 for the cleaned estimator follows the shrinkage-target notation of the companion paper [21]; it is unrelated to the sample size T.

2.1 Primal-dual active-set reduction

The budget constraint $\mathbf{1}^\top w = 1$ and non-negativity $w \geq 0$ are handled separately. Fix a set $\mathcal{A} \subseteq \{1, \ldots, n\}$ of active assets. The budget-constrained subproblem on $\mathcal{A}$,

$$\min_{w_{\mathcal{A}}:\, \mathbf{1}^\top w_{\mathcal{A}}=1} \; w_{\mathcal{A}}^\top T_{0,\mathcal{A}}^{\mathrm{RMT}}\, w_{\mathcal{A}} - \rho\, \hat{\mu}_{\mathcal{A}}^\top w_{\mathcal{A}}\,,$$

has, by the KKT conditions, the closed-form solution

$$w_{\mathcal{A}}^\star \;=\; \theta\, v_1 + \tfrac{\rho}{2} v_2\,, \qquad v_1 = (T_{0,\mathcal{A}}^{\mathrm{RMT}})^{-1}\mathbf{1}\,, \quad v_2 = (T_{0,\mathcal{A}}^{\mathrm{RMT}})^{-1}\hat{\mu}_{\mathcal{A}}\,,$$

where the budget multiplier $\theta = (1 - \frac{\rho}{2}\,\mathbf{1}^\top v_2)/(\mathbf{1}^\top v_1)$ is recovered analytically from $\mathbf{1}^\top w_{\mathcal{A}}^\star = 1$.

Non-negativity is enforced by the following primal-dual outer loop. Throughout, the objective gradient is

$$g \;=\; 2\, T_0^{\mathrm{RMT}} w - \rho\, \hat{\mu}\,,$$

evaluated through the low-rank form $T_0^{\mathrm{RMT}} w = \bar{\lambda} w + U_k(\Delta_k(U_k^\top w))$ at cost $O(nk)$ (Proposition 3.1); the return matrix X never enters the solve. Stationarity of the restricted subproblem implies $g_j = 2\theta$ for every $j \in \mathcal{A}$, so the common active-set gradient value recovers the multiplier exactly; the implementation uses the median over $\mathcal{A}$, which is exact up to roundoff.

- *Primal step*: if $w_{\mathcal{A},i} < -\varepsilon$ for some $i \in \mathcal{A}$, remove the most negative such asset and re-solve.
- *Dual step*: compute g_i for $i \notin \mathcal{A}$ and let $\nu = \mathrm{median}_{j\in\mathcal{A}}(g_j)$. If $g_i < \nu - \varepsilon$ for some excluded asset, add the most violated asset to $\mathcal{A}$ and re-solve.
- *Termination*: stop when $w_{\mathcal{A},i} \geq -\varepsilon$ for all $i \in \mathcal{A}$ and $g_i \geq \nu - \varepsilon$ for all $i \notin \mathcal{A}$.

Proposition 2.1 (Finite termination)**.** *Under generic non-degeneracy of the KKT conditions (no two primal weights are simultaneously at zero; no two excluded assets are tied for the largest dual violation), the primal-dual outer loop terminates in finitely many iterations.*

Proof. Since $T_{0,\mathcal{A}}^{\mathrm{RMT}} = \bar{\lambda} I_{n_a} + U_{k,\mathcal{A}} \Delta_k U_{k,\mathcal{A}}^\top$ with $\bar{\lambda} > 0$ and $\delta_j > 0$ for all j, we have $T_{0,\mathcal{A}}^{\mathrm{RMT}} \succ 0$ for every $\mathcal{A}$, so the restricted subproblem has a unique optimum. Each outer step exchanges a single asset. Under the non-degeneracy assumption the restricted dual objective strictly decreases at each complete primal-dual exchange by the standard parametric QP pivot argument [18]; consequently no active set is visited twice. Since the 2^n subsets of $\{1, \ldots, n\}$ are finite, the algorithm terminates. ☐

Remark 2.2 (Implementation safeguards)**.** *Non-degeneracy is generic but cannot be verified a priori. The reference implementation therefore carries two safeguards: an iteration cap, and a check that the active set changed between consecutive outer steps. As an acceleration, assets whose weights are strongly negative ($w_i < -10\,\varepsilon$) are dropped in bulk before the single-exchange rule applies; in all experiments of Section 7 the safeguards never triggered and bulk drops did not affect the returned portfolio (the converged weights agree with interior-point solutions to solver accuracy, Section 7.1).*

Remark 2.3 (Tolerance scale)**.** *The tolerance ε applies to two quantities with different units: portfolio weights (primal step) and gradient entries (dual step). Weights are scale-free; gradient entries scale with $\bar{\lambda}$, which is of order 10^{-4} for daily returns. The default $\varepsilon = 10^{-6}$ is appropriate for daily return data; for data on other scales the dual tolerance should be taken relative to $\bar{\lambda}$.*

This paper develops the inner solve: when $T_0 = T_0^{\mathrm{RMT}}$, the inverse $(T_{0,\mathcal{A}}^{\mathrm{RMT}})^{-1}$ is available in closed form via the Woodbury identity (Section 4), reducing the dominant cost per outer step from $O(n_a^2 k + n_a^3)$ to $O(n_a k^2 + k^3)$.

3 The RMT-Cleaned Covariance Estimator

3.1 Marchenko–Pastur spectral separation

The Marchenko–Pastur (MP) law [15] describes the limiting spectral distribution of $\hat{\Sigma} = X^\top X/T$ when X has i.i.d. entries with variance σ^2 and $n, T \to \infty$ with $n/T \to c \in (0, 1)$. Under the null hypothesis that $\hat{\Sigma}$ is a rescaled Wishart matrix (no true signal), the bulk of the empirical spectral distribution (ESD) is supported on $[\lambda_-, \lambda_+]$ where

$$\lambda_\pm = \sigma^2(1 \pm \sqrt{c})^2 .$$

In finite samples, σ^2 is estimated by the average entry-level variance $\hat{\sigma}^2 = \|X\|_F^2/(nT) = \mathrm{tr}(\hat{\Sigma})/n$ (the grand mean eigenvalue) and c is replaced by n/T, giving the empirical edges $\hat{\lambda}_\pm = \hat{\sigma}^2(1 \pm \sqrt{n/T})^2$.

Eigenvalues above $\hat{\lambda}_+$ are not explained by the null and are attributed to genuine covariance structure. Eigenvalues below $\hat{\lambda}_+$ are consistent with noise and carry no reliable directional information. We denote by k the number of signal eigenvalues: $k = |\{j : \lambda_j(\hat{\Sigma}) > \hat{\lambda}_+\}|$. For S&P 500 data with $n = 494$ and $T = 1213$, $\hat{\lambda}_+ \approx 1.22 \times 10^{-3}$ and $k = 23$; for the factor-calibrated synthetic model of Section 7, the detected k grows approximately as $n/21$ in this regime.

3.2 Construction of T_0^{RMT}

Let $\hat{\Sigma} = U\Lambda U^\top$ be the eigendecomposition of the sample covariance. Partition the eigenpairs by the MP edge: let $U_k \in \mathbb{R}^{n\times k}$ hold the k signal eigenvectors with eigenvalues $\Lambda_k = \mathrm{diag}(\lambda_1, \ldots, \lambda_k)$, and let $U_0 \in \mathbb{R}^{n\times(n-k)}$ hold the noise eigenvectors. The bulk eigenvalues are replaced by their average,

$$\bar{\lambda} \;=\; \frac{1}{n-k}\sum_{j>k}\lambda_j \;=\; \frac{\mathrm{tr}(\hat{\Sigma}) - \sum_{j\le k}\lambda_j}{n-k} ,$$

which preserves the trace – total variance – of the sample covariance. The resolution of the identity $U_kU_k^\top + U_0U_0^\top = I$ gives

$$T_0^{\mathrm{RMT}} = U_k\Lambda_kU_k^\top + \bar{\lambda}\,U_0U_0^\top = \bar{\lambda}I + U_k\Delta_kU_k^\top ,$$

where $\Delta_k = \Lambda_k - \bar{\lambda}I = \mathrm{diag}(\delta_1, \ldots, \delta_k)$ with $\delta_j = \lambda_j - \bar{\lambda} > 0$ for all j (since $\lambda_j > \hat{\lambda}_+ > \hat{\sigma}^2 > \bar{\lambda}$). This is exactly the "constant residual eigenvalue" (CRE) – or eigenvalue clipping – estimator introduced by Laloux et al. [9], studied in depth by Bun, Bouchaud and Potters [3], and widely adopted in quantitative finance following López de Prado [14].

Proposition 3.1 (Spectral properties of T_0^{RMT})**.** *The RMT-cleaned estimator satisfies:*

1. $T_0^{\mathrm{RMT}} \succ 0$ *with eigenvalues* $\{\lambda_1, \ldots, \lambda_k, \bar{\lambda}, \ldots, \bar{\lambda}\}$*, and* $\mathrm{tr}(T_0^{\mathrm{RMT}}) = \mathrm{tr}(\hat{\Sigma})$*.*
2. $\kappa(T_0^{\mathrm{RMT}}) = \lambda_{\max}(\hat{\Sigma})/\bar{\lambda}$*.*
3. $T_0^{\mathrm{RMT}} = \hat{\Sigma} - U_0(\Lambda_0 - \bar{\lambda}I)U_0^\top$*, i.e.,* T_0^{RMT} *differs from* $\hat{\Sigma}$ *only in the noise subspace.*
4. *The matrix-vector product* $T_0^{\mathrm{RMT}}v = \bar{\lambda}v + U_k(\Delta_k(U_k^\top v))$ *costs* $O(nk)$ *and requires* $O(nk)$ *storage.*

Proof. (1) follows from the eigendecomposition $T_0^{\mathrm{RMT}} = [U_k\ U_0]\,\mathrm{diag}(\Lambda_k, \bar{\lambda}I_{n-k})\,[U_k\ U_0]^\top$, $\delta_j > 0$ for all signal eigenpairs, and the definition of $\bar{\lambda}$ as the bulk average. (2) follows from (1): the maximum eigenvalue is $\lambda_1 > 0$ and the minimum is $\bar{\lambda} > 0$. (3) is direct algebra. (4) The product costs two dense matrix-vector products with $U_k \in \mathbb{R}^{n\times k}$. □

Remark 3.1 (Condition number)**.** T_0^{RMT} *achieves* $\kappa = \lambda_{\max}(\hat{\Sigma})/\bar{\lambda}$*: noise eigenvalues are lifted to the floor* $\bar{\lambda}$*, signal eigenvalues are preserved. On S&P 500 data this gives* $\kappa \approx 343$*, versus* $\kappa(\hat{\Sigma}) \approx 106{,}000$ *– an improvement by a factor of roughly* 300*. The low condition number is a consequence of replacing the* $(n-k)$*-dimensional noise subspace eigenvalue spectrum (which spans several decades) with a single value* $\bar{\lambda}$*; it also keeps the* $k \times k$ *Woodbury correction matrix* W *well-conditioned regardless of* n*.*

Remark 3.2 (Relation to optimal RMT cleaning)**.** *CRE clipping is the simplest member of the family of rotationally invariant estimators (RIE). It is not loss-optimal: the oracle RIE replaces* every *eigenvalue – bulk and signal alike – by a value determined by the eigenvector overlaps of Ledoit and Péché [12], and these oracle values are not constant across the bulk; nonlinear shrinkage [13, 11] and the RIE of Bun et al. [3] dominate clipping in Frobenius loss. In particular, CRE retains the signal eigenvalues unshrunk although eigenvalues just above the edge are biased upward and their eigenvectors imperfectly aligned with the population factors (the BBP transition [1, 8]). The reason this paper nevertheless works with CRE is structural: clipping is the member of the family whose output is* low-rank-plus-identity *(a spiked structure in the sense of Johnstone [8]), and only that structure admits the closed-form Woodbury solve of Section 4. Nonlinear shrinkage and general RIE produce full-rank spectral modifications to which the Woodbury identity does not apply; portfolios built on them require a general factorisation. Whether the statistical gap matters for long-only minimum-variance portfolios is an empirical question; on our data it is small (Section 7.5).*

Remark 3.3 (Externally specified factor models)**.** *The MP threshold provides a data-driven rule for selecting* k*, but the Woodbury solver applies for any choice of* k*. Practitioners with an external factor model – a commercial risk model with pre-specified style and industry factors, or a PCA with* k *fixed by cross-validation – can set* $k = k_{\mathrm{ext}}$ *and supply the corresponding eigenpairs directly, bypassing the MP threshold entirely. The key structural assumption for the Woodbury solve is not the source of* k *but the replacement of the remaining* $n-k$ *eigenvalues by a single scalar* $\bar{\lambda}$*. The rSVD preprocessing of Section 5 and the complexity bound of Proposition 4.3 hold with* k *replaced by* k_{ext}*. When* k_{ext} *is known in advance, the preprocessing cost can be reduced to* $O(nTk_{\mathrm{ext}})$ *by targeting only the required number of components rather than applying the MP threshold post-hoc.*

3.3 Choice of k

The threshold k is determined by the MP upper edge $\hat{\lambda}_+$. In practice k is stable across the S&P 500 sample: varying the estimation window over the five-year period gives $k \in [21, 25]$, with the central value $k = 23$. An audit at $k-1$, k, and $k+1$ adds negligible cost (one extra Woodbury solve per audit point) and reveals the portfolio's sensitivity to the threshold, which is small in the bulk of the frontier (Section 6).

Two systematic biases in k deserve mention.

Finite-sample edge fluctuations. At finite T the largest noise eigenvalue fluctuates around the bulk edge at the Tracy–Widom scale $O(T^{-2/3})$ [8], so a small number of noise eigenvalues may be classified as signal (or vice versa); refined spectrum estimators such as El Karoui's [4] reduce this effect. For the sample sizes used here ($T \geq 250$) the impact on k is at most one or two eigenvalues, which is precisely the perturbation covered by the $k \pm 1$ audit of Section 6.

Sparse structure below the detection threshold. Sector correlations below the MP edge contribute a diffuse signal that the threshold does not detect. If these directions are material for the portfolio, the audit at $k \pm 1$ will show sensitivity; in practice, the minimum-variance portfolio is insensitive to directions with small eigenvalues because those directions contribute little variance.

4 The Woodbury Portfolio Algorithm

4.1 Woodbury direct solve

The system matrix on the active set $\mathcal{A}$ is $T_{0,\mathcal{A}}^{\mathrm{RMT}} = \bar{\lambda} I_{n_a} + U_{k,\mathcal{A}} \Delta_k U_{k,\mathcal{A}}^\top$, where $U_{k,\mathcal{A}} = U_k[\mathcal{A}, :] \in \mathbb{R}^{n_a \times k}$ is the restriction of the signal eigenvectors to the active assets and $n_a = |\mathcal{A}|$. The Woodbury matrix identity [5], applied with $A = \bar{\lambda} I_{n_a}$, $U = U_{k,\mathcal{A}}$, $C = \Delta_k$, $V = U_{k,\mathcal{A}}^\top$, gives

$$\big(T_{0,\mathcal{A}}^{\mathrm{RMT}}\big)^{-1} b \;=\; \frac{b}{\bar{\lambda}} \;-\; \frac{1}{\bar{\lambda}^2} U_{k,\mathcal{A}} W^{-1} U_{k,\mathcal{A}}^\top b, \qquad W = \Delta_k^{-1} + \frac{1}{\bar{\lambda}} U_{k,\mathcal{A}}^\top U_{k,\mathcal{A}} \;\in \mathbb{R}^{k \times k}.$$

Note that after restricting to $\mathcal{A}$ the columns of $U_{k,\mathcal{A}}$ are generally no longer orthonormal: $U_{k,\mathcal{A}}^\top U_{k,\mathcal{A}} \neq I_k$. The correction matrix W accounts for this; its Cholesky factorisation is used to evaluate $W^{-1} U_{k,\mathcal{A}}^\top b$.

Proposition 4.1 (Woodbury solve complexity)**.** *One application of $(T_{0,\mathcal{A}}^{\mathrm{RMT}})^{-1}$ costs:*

- *$O(n_a k)$ for the two matrix-vector products $U_{k,\mathcal{A}}^\top b$ and $U_{k,\mathcal{A}} z$;*
- *$O(n_a k^2)$ to form $W = \Delta_k^{-1} + \bar{\lambda}^{-1} U_{k,\mathcal{A}}^\top U_{k,\mathcal{A}}$ (a rank-k outer product sum over n_a columns);*
- *$O(k^3)$ for the Cholesky factorisation of W.*

The dominant term is $O(n_a k^2)$; the total per-application cost is $O(n_a k^2 + k^3)$.

4.2 Algorithm and warm-starting

The tolerance $\varepsilon = 10^{-6}$ governs both the primal feasibility check (line 7: drop assets with $w < -\varepsilon$) and the dual violation threshold (line 12: admit assets with $g_i < \nu - \varepsilon$); see Remark 2.3 for its scale. This is a standard KKT tolerance for active-set methods [18].

Two further features of Algorithm 1 deserve emphasis.

The return matrix never enters the solve. The dual-feasibility check at line 10 evaluates the gradient of the objective, $g = 2\, T_0^{\mathrm{RMT}} w - \rho \hat{\mu}$, through the low-rank form at cost $O(nk)$. Neither $\hat{\Sigma}$ nor X is touched after preprocessing: the entire solve operates on the $O(nk)$ factors $(U_k, \Delta_k, \bar{\lambda})$.

Warm-starting profits from active-set continuity, not initial guesses. For a direct solver there is no initial guess to warm-start; the Woodbury solve does not iterate. Warm-starting passes the previous active-set $\mathcal{A}$ and weight vector w, reducing the number of outer primal-dual steps. On the efficient frontier sweep with $\rho \in [0, \rho_{\max}]$ the active set changes by at most a few assets between adjacent ρ values, so the warm solver typically terminates in one to two outer steps.

This continuity is a structural property of parametric quadratic programming, not an artefact of the data-generating process. The Markowitz efficient frontier is a continuous piecewise-linear curve in weight space as a function of ρ; at each breakpoint, generically, exactly one asset enters or leaves the active set [16]. Between breakpoints the active set is constant. Choosing a fine enough ρ grid – e.g. 50 equally spaced points – ensures that adjacent solves share all but a few active assets regardless of whether the returns are synthetic or real, and regardless of the covariance estimator used.

Remark 4.1 (Relation to the critical line algorithm)**.** *The same piecewise-linear structure underlies the critical line algorithm (CLA) [16, 17, 22], which traces the exact frontier by pivoting from breakpoint to breakpoint instead of re-solving on a ρ grid. On a fixed active set the restricted solution is affine in ρ, $w_{\mathcal{A}}(\rho) = a + \rho b$ with $a = v_1/(\mathbf{1}^\top v_1)$ and $b = \frac{1}{2}\big(v_2 - \frac{\mathbf{1}^\top v_2}{\mathbf{1}^\top v_1} v_1\big)$, and the multiplier*

Algorithm 1 Woodbury portfolio solver

Require: Signal eigenpairs $(U_k, \Lambda_k) \in \mathbb{R}^{n\times k} \times \mathbb{R}^k$; bulk mean $\bar{\lambda}$; return-tilt $\rho \geq 0$; expected returns $\hat{\mu}$; tolerance $\varepsilon = 10^{-6}$; optional warm start $(\mathcal{A}_0, w_0)$

Ensure: Portfolio weights $w \in \mathbb{R}^n$ with $w \geq 0$, $\mathbf{1}^\top w = 1$

$\mathcal{A} \leftarrow \mathcal{A}_0$ (or $\{1, \dots, n\}$ for cold start); $w \leftarrow w_0$ (or $\mathbf{0}$); $\Delta_k \leftarrow \Lambda_k - \bar{\lambda}\mathbf{1}$
loop
    $U_a \leftarrow U_k[\mathcal{A}, :]$ ▷ $n_a \times k$ restriction
    $W \leftarrow \operatorname{diag}(1/\Delta_k) + U_a^\top U_a/\bar{\lambda}$ *($O(n_a k^2)$)*
    $L \leftarrow \operatorname{chol}(W)$; define $\operatorname{inv}(b) \leftarrow b/\bar{\lambda} - U_a\,(L^\top \backslash (L \backslash U_a^\top b))/\bar{\lambda}^2$ *($O(k^3)$ once)*
    $v_1 \leftarrow \operatorname{inv}(\mathbf{1}_{n_a})$; $v_2 \leftarrow \operatorname{inv}(\hat{\mu}_{\mathcal{A}})$ if $\rho > 0$ *($O(n_a k)$ each)*
    Recover $\theta = (1 - \frac{\rho}{2}\mathbf{1}^\top v_2)/(\mathbf{1}^\top v_1)$; $w_{\mathcal{A}} = \theta v_1 + \frac{\rho}{2} v_2$
    if $\exists\, i \in \mathcal{A}$ with $w_{\mathcal{A},i} < -\varepsilon$ **then**
        $\mathcal{A} \leftarrow \mathcal{A} \setminus \{\arg\min_i w_{\mathcal{A},i}\}$ *(primal step; cf. Remark 2.2)*
    else
        $w \leftarrow 0$; $w[\mathcal{A}] \leftarrow w_{\mathcal{A}}$
        $g \leftarrow 2\big(\bar{\lambda} w + U_k(\Delta_k(U_k^\top w))\big) - \rho\hat{\mu}$; $\nu \leftarrow \operatorname{median}_{j\in\mathcal{A}}(g_j)$ *($O(nk)$)*
        $j^\star \leftarrow \arg\min_{i \notin \mathcal{A}} g_i$
        if $g_{j^\star} \geq \nu - \varepsilon$ **then break** *(KKT satisfied)*
        end if
        $\mathcal{A} \leftarrow \mathcal{A} \cup \{j^\star\}$ *(dual step)*
    end if
end loop
return w

$\nu(\rho) = 2\theta(\rho)$ is affine as well, so every primal exit (w_i reaching zero) and dual entry ($g_i - \nu$ reaching zero) is the root of an affine function: the exact frontier costs one Woodbury solve per breakpoint. We implement this sweep with the Woodbury kernel as its linear algebra and validate it against the grid sweep in Section 7.4. The two approaches remain complementary in practice: CLA enumerates the exact frontier for a fixed *covariance, whereas the warm-started grid sweep extends unchanged to the practitioner's setting in which consecutive solves differ in the data (a refreshed covariance at each rebalance) rather than only in ρ.*

Remark 4.2 (Ridge terms and other structured modifications)**.** *The Woodbury structure survives several practically relevant modifications. A ridge penalty $\gamma\|w\|^2$ simply shifts the identity coefficient, $\bar{\lambda} \mapsto \bar{\lambda} + \gamma$. A diagonal loading $D + U_k\Delta_k U_k^\top$ with non-scalar $D \succ 0$ replaces $b/\bar{\lambda}$ by $D^{-1}b$ at unchanged cost. Additional* linear equality *constraints (e.g. sector-neutrality) enlarge the KKT system by one row per constraint; the inverse of the leading block is still available through Woodbury, so the solve cost grows only linearly in the number of constraints. What is* not *supported is a full-rank perturbation of the bulk spectrum, as produced by nonlinear shrinkage (Remark 3.2).*

4.3 Complexity and crossover analysis

Proposition 4.2 (Woodbury vs. KKT-Cholesky crossover)**.** *Let the KKT-Cholesky inner solve assemble $T_{0,\mathcal{A}}^{\mathrm{RMT}}$ explicitly via a rank-k symmetric update ($n_a^2 k$ flops to leading order) and factor by Cholesky ($n_a^3/3$ flops), and let the Woodbury inner solve form W ($n_a k^2$ flops) and factor it ($k^3/3$ flops). Then, writing $x = k/n_a$, the Woodbury solve requires fewer flops per outer step if and only if*

$$n_a k^2 + \tfrac{k^3}{3} \;<\; n_a^2 k + \tfrac{n_a^3}{3} \iff x^3 + 3x^2 - 3x - 1 < 0 \iff (x-1)\,(x^2 + 4x + 1) < 0 \iff x < 1\,,$$

i.e. exactly when $k < n_a$. The same threshold holds with unit constants: $n_a k^2 + k^3 < n_a^2 k + n_a^3 \iff (x+1)^2(x-1) < 0 \iff x < 1$.

Proof. Divide both sides by n_a^3 and substitute $x = k/n_a$; the cubics factor as displayed, and for $x > 0$ the second factor is positive in both cases. □

Remark 4.3 (Crossover in practice). *The crossover condition $k < n_a$ is satisfied throughout our experiments: on S&P 500 data $k = 23$ against $n_a = 63$ at the minimum-variance solution ($x \approx 0.37$), and on the synthetic frontier x ranges from about 0.7 at the tightest point ($n_a = 31$) to about 0.2 at the high-return end ($n_a = 97$). At $x \approx 0.5$ the flop-count advantage predicted by Proposition 4.2 is about 3×; the measured per-step advantage is larger (5× warm, Table 3), which is consistent with the Woodbury working set ($k \times k$) fitting in lower cache levels than the assembled $n_a \times n_a$ system, though we have not instrumented the memory hierarchy directly. When $n_a \le k$ – which does not occur for long-only equity portfolios in our experiments – the solver should fall back to the KKT-Cholesky branch; the Woodbury solve itself remains valid for all $n_a \ge 1$, since $W = \Delta_k^{-1} + \bar{\lambda}^{-1} U_{k,\mathcal{A}}^\top U_{k,\mathcal{A}}$ is the sum of a positive definite diagonal matrix and a positive semidefinite matrix, hence always invertible.*

Proposition 4.3 (Per-solve pipeline complexity). *Let s denote the number of outer active-set steps for one portfolio solve. The solve cost is*

$$C_{\text{Woodbury}}(s) \;=\; s \cdot O(n_a k^2 + k^3 + nk)\,,$$

where the $O(nk)$ term is the gradient evaluation of the dual step. For a sweep of S solves sharing the same eigenpairs (U_k, Λ_k), the total pipeline cost including preprocessing is

$$C_{\text{total}} \;=\; \underbrace{O(nTk)}_{\textit{preprocessing}} + \underbrace{S \cdot s \cdot O(n_a k^2 + k^3 + nk)}_{\textit{solves}}\,,$$

and the preprocessing cost amortises to $O(nTk/S)$ per solve.

Proof. Preprocessing via randomised SVD costs $O(nT(k+p))$ for oversampling parameter p (Proposition 5.1); setting $p = O(k)$ gives $O(nTk)$. Each outer step costs one Woodbury application (Proposition 4.1) plus one $O(nk)$ gradient evaluation. The pipeline cost is the sum; the amortisation claim is division by S. □

At the $n = 500$ benchmark the measured preprocessing share of the total pipeline for a 50-point frontier sweep is between roughly 30% (cold sweep) and 45% (warm sweep); for a single solve the preprocessing dominates (Table 4). In every case the pipeline total remains well below the cost of the KKT-Cholesky solve alone at large n (Section 7.2).

5 Randomised SVD Preprocessing

The Woodbury solver requires the top-k eigenpairs of $\hat{\Sigma} = X^\top X/T$. Two paths are available.

Dense path Assemble $\hat{\Sigma} = X^\top X/T$ explicitly ($O(n^2T)$ FLOPs, $O(n^2)$ storage), then compute a full eigendecomposition ($O(n^3)$). Total cost: $O(n^2T + n^3)$. At $n = 494$, $T = 1213$ this is approximately 3×10^8 FLOPs and 1.9×10^6 doubles (about 15 MB). Practical: 0.013 s on the benchmark hardware (Table 1).

Randomised SVD path The right singular vectors of $X/\sqrt{T}$ are the eigenvectors of $\hat{\Sigma}$. A randomised truncated SVD [6] computes the top-k eigenpairs without forming $X^\top X$. The trace-preserving bulk mean is recovered without any dense decomposition from $\bar{\lambda} = \big(\mathrm{tr}(\hat{\Sigma}) - \sum_{j\le k} \hat{\lambda}_j\big)/(n-k)$ with $\mathrm{tr}(\hat{\Sigma}) = \|X\|_F^2/T$.

Algorithm 2 Randomised truncated SVD for eigenpairs of $\hat{\Sigma}$

Require: Returns $X \in \mathbb{R}^{T\times n}$; target rank k; oversampling $p \ge 5$
Ensure: Signal eigenvectors $U_k \in \mathbb{R}^{n\times k}$, eigenvalues $\Lambda_k \in \mathbb{R}^k$, bulk mean $\bar{\lambda}$

1: $\Omega \leftarrow \mathcal{N}(0, I_{n\times(k+p)})$ ▷ random test matrix
2: $Y \leftarrow X^\top(X\Omega)$ ▷ $n \times (k+p)$; two matrix-vector products with X, $O(nT(k+p))$
3: $Q \leftarrow \mathrm{orth}(Y)$ ▷ orthonormal basis, $O(n(k+p)^2)$
4: $B \leftarrow (XQ)^\top(XQ)/T$ ▷ $(k+p)\times(k+p)$ projected covariance; no $X^\top X$ formed
5: $[\hat{U}, \hat{\Lambda}, \sim] \leftarrow \mathrm{eigh}(B)$; sort descending
6: $U_k \leftarrow Q\hat{U}[:, 1{:}k]$; $\Lambda_k \leftarrow \hat{\Lambda}[1{:}k]$
7: $\bar{\lambda} \leftarrow \big(\|X\|_F^2/T - \sum_{j\le k}\hat{\Lambda}_j\big)/(n-k)$; identify k via MP threshold
8: **return** $(U_k, \Lambda_k, \bar{\lambda})$

Proposition 5.1 (Randomised SVD cost and accuracy). *Algorithm 2 costs $O(nT(k+p))$ FLOPs and $O(n(k+p))$ storage; no $n\times n$ matrix is formed. Let $P = QQ^\top$ denote the orthogonal projector onto the computed basis. Since the sketch $Y = T\,\hat{\Sigma}\,\Omega$ applies the range finder of [6] to the symmetric matrix $\hat{\Sigma}$ with Gaussian test matrix Ω and oversampling $p \ge 2$, Theorem 10.6 of [6] gives, in expectation,*

$$\mathbb{E}\,\big\|(I-P)\hat{\Sigma}\big\|_2 \;\le\; \Big(1+\sqrt{\tfrac{k}{p-1}}\Big)\,\lambda_{k+1} \;+\; \frac{e\sqrt{k+p}}{p}\Big(\textstyle\sum_{j>k}\lambda_j^2\Big)^{1/2} \;=:\; \epsilon_{k,p}\,,$$

where λ_j are the eigenvalues of $\hat{\Sigma}$. The returned eigenpairs are the exact eigenpairs of the compression $P\hat{\Sigma}P$, and by Weyl's inequality the eigenvalue errors satisfy $|\lambda_j - \hat{\lambda}_j| \le \|\hat{\Sigma} - P\hat{\Sigma}P\|_2 \le 2\,\|(I-P)\hat{\Sigma}\|_2$ for all $j \le k$.

Proof. The cost and storage counts follow from the line-by-line annotations of Algorithm 2. The error bound is Theorem 10.6 of [6] applied to $A = \hat{\Sigma}$ (the deterministic scaling T of the sketch does not change its range). For the compression bound, write $\hat{\Sigma} - P\hat{\Sigma}P = (I-P)\hat{\Sigma} + P\hat{\Sigma}(I-P)$ and note that both terms have spectral norm at most $\|(I-P)\hat{\Sigma}\|_2$ by symmetry; Weyl's inequality [7] then bounds each eigenvalue perturbation by the norm of the difference. □

The bound $\epsilon_{k,p}$ is governed by the eigenvalues at and below the MP edge: $\lambda_{k+1} \approx \hat{\lambda}_+ \approx 1.2\times10^{-3}$ on S&P 500 data, while the leading signal eigenvalue $\lambda_1 \approx 343\,\bar{\lambda} \approx 0.066$ is roughly 50× larger. The bound is conservative – the tail-sum term accumulates all $n-k$ bulk eigenvalues – and the measured accuracy is far better, as the direct portfolio comparison below shows.

Memory passes Algorithm 2 involves four matrix-vector products with X in total: two for the sketch $Y = X^\top(X\Omega)$ (line 2) and two for the projection $B = Q^\top X^\top XQ$ (line 4, computed as $(XQ)^\top(XQ)/T$). Reading X from memory is required at two distinct stages: once for the sketch (line 2) and once for the projection (line 4). For very large n and T where X does not fit in memory, a single-pass variant that combines both stages is available at a modest increase in approximation error; see [6] for details. At benchmark size ($n = 494$, $T = 1213$, approximately 4.6 MB for double-precision X) both stages fit comfortably in L3 cache.

Method	Time (s)	Storage	Subspace error
Dense eigendecomposition	0.0127	n^2	—
Randomised SVD ($p = 10$)	0.0069	nk	$5.7e - 02$
Speedup	1.8×	21×	

Table 1: Preprocessing paths for S&P 500 ($n = 494$, $T = 1213$, $k = 23$, $p = 10$). Subspace error: $\|UU^\top - \hat{U}\hat{U}^\top\|_F$, measuring alignment of the top-k signal subspaces. The randomised SVD is 1.8× faster and requires 21× less storage.

Benchmark The randomised SVD is 1.8× faster on the benchmark hardware at $n = 494$. The advantage grows with n: at $n = 3000$ (Table 4) the dense path takes 1.25 s versus 0.05 s for the randomised path, a 25× wall-clock advantage. The subspace error of 5.7×10^{-2} is the Frobenius norm $\|UU^\top - \hat{U}\hat{U}^\top\|_F$; relative to the subspace scale $\sqrt{k} \approx 4.8$ this is approximately 1.2% alignment error per eigenvector column. The portfolio impact is measured directly: solving the minimum-variance problem with the rSVD eigenpairs versus the dense-path eigenpairs (same $k = 23$, $\alpha = 1$) changes no portfolio weight by more than 2.3 basis points. For comparison, adding or removing a *complete* eigenvector from the signal subspace (the $k \pm 1$ audit of Section 6) moves weights by up to 33 basis points; the preprocessing approximation is an order of magnitude below the threshold uncertainty that any user of the MP rule already accepts.

Large-n recommendation For $n \leq 500$ either path is practical; the dense path is simpler to implement and both finish in milliseconds. For $n > 1000$ the randomised SVD is strongly preferred: the dense path's $O(n^2)$ storage requirement alone (8 MB at $n = 1000$, 72 MB at $n = 3000$) begins to stress L3 cache, while the randomised path's $O(n(k + p))$ requirement (about 3 MB at $n = 3000$, $k = 123$) stays within typical L3.

Incremental updates For daily rebalancing, X changes by one row (adding the new day, dropping the oldest), so $\hat{\Sigma}$ changes by a rank-two update. We implement an incremental variant that maintains the tracked basis $Q \in \mathbb{R}^{n \times m}$ ($m = k + p$) and the compressed eigenvalues across days: the basis is expanded by the orthogonal residuals of the entering and leaving observations, the $(m+2) \times (m+2)$ projected matrix is re-diagonalised, and the top m directions are retained, at cost $O(nm^2 + m^3)$ per day with no re-sketch of X (the mean update, a further rank-one correction, is omitted; daily means are negligible relative to volatilities). On the S&P 500 benchmark ($m = 33$) one update costs 0.21 ms against 4.8 ms for a fresh randomised SVD of the window – a 23× speedup. Tracking accuracy degrades slowly: over 100 consecutive daily updates with no re-sketch, the minimum-variance portfolio built from the tracked eigenpairs differs from the freshly-sketched one by 6.7 bp in the median day and 38 bp in the worst, i.e. the worst-case drift after five months reaches the same order as the $k \pm 1$ threshold uncertainty. A monthly re-sketch (amortised cost 0.2 ms/day) keeps the drift an order of magnitude below it.

6 Threshold Robustness and k-Sensitivity

The Marchenko–Pastur threshold determines k from data. A practical concern is whether the portfolio is sensitive to this choice: a threshold that incorrectly includes one extra noise factor (or excludes one signal factor) by a small amount would be problematic if the resulting portfolio changes substantially.

6.1 Cost of auditing

Auditing the sensitivity at $k \pm 1$ requires two additional Woodbury solves. Since the eigenpairs (U_k, Λ_k) are already computed, these are pure solve costs at $O(n_a k^2 + k^3)$ each – negligible relative to the preprocessing. On the benchmark dataset, the three-point audit ($k = 22, 23, 24$) adds less than 3 ms to the total pipeline. This makes the audit a costless safety check.

6.2 Analytical sensitivity

Let $w^*(k)$ denote the RMT minimum-variance portfolio with k signal factors. To leading order – ignoring the accompanying $O(1/(n-k))$ adjustment of the bulk mean $\bar{\lambda}$, which is at the percent level on the benchmark data – adding the $(k+1)$-th factor changes the system matrix by a rank-one update: $T_0^{\mathrm{RMT}}(k+1) \approx T_0^{\mathrm{RMT}}(k) + \delta_{k+1}\, u_{k+1} u_{k+1}^\top$, where $\delta_{k+1} = \lambda_{k+1} - \bar{\lambda} > 0$. By the Sherman–Morrison identity,

$$(T_0^{\mathrm{RMT}}(k+1))^{-1} \approx (T_0^{\mathrm{RMT}}(k))^{-1} - \frac{\delta_{k+1}}{1 + \delta_{k+1} u_{k+1}^\top (T_0^{\mathrm{RMT}}(k))^{-1} u_{k+1}} \cdot (T_0^{\mathrm{RMT}}(k))^{-1} u_{k+1} u_{k+1}^\top (T_0^{\mathrm{RMT}}(k))^{-1}.$$

The portfolio perturbation $w^*(k+1) - w^*(k)$ is therefore controlled by δ_{k+1}, the eigenvalue excess of the marginal factor. Since λ_{k+1} lies just above $\hat{\lambda}_+$ by construction, δ_{k+1} is small relative to the signal excesses $\delta_1, \ldots, \delta_k$, and the portfolio perturbation is correspondingly small. The empirical audit below includes the exact bulk-mean adjustment.

6.3 Empirical sensitivity on S&P 500 data

Table 2 quantifies the change for $k \in \{22, 23, 24\}$ on the full S&P 500 sample ($n = 494$, $T = 1213$). The maximum absolute weight change across all assets is 33 bp when one signal factor is removed ($k = 22$) and 8 bp when one is added ($k = 24$); the ℓ_2 weight-vector distances are 95 bp and 27 bp respectively. The annualised volatility of the minimum-variance portfolio changes by less than one basis point in either direction.

k	Active	$\mathrm{Vol}_{\mathrm{ann}}$ (%)	$\|w(k) - w(k^*)\|_\infty$	$\|w(k) - w(k^*)\|_2$
22	96	10.53	0.0012	0.0037
23	95	10.52	0.0000	0.0000
24	95	10.52	0.0001	0.0003

Table 2: Portfolio sensitivity to k at $\rho = 0$ on S&P 500 data ($n = 494$, $T = 1213$). $k^* = 23$ is the MP threshold. All three Woodbury solves complete in under 3 ms combined (preprocessing amortised). Maximum absolute weight change: 33 bp.

These results indicate that the MP threshold is not a sensitive hyperparameter for this problem: misjudging k by one factor moves individual weights by a few tens of basis points and the portfolio volatility by less than one basis point. A practitioner who is uncertain whether the correct k is 22, 23, or 24 can audit all three at negligible cost.

7 Numerical Results

All experiments were run on an Apple M4 Pro (14-core CPU, 48 GB RAM) under Python 3.12 with NumPy 2.4 (Apple Accelerate BLAS), SciPy 1.17, and scikit-learn 1.x. Reported times are the minimum of three repeated runs, the standard protocol for suppressing scheduling noise; the spread across repetitions was below ten percent for all entries. The solver comparison and scaling experiments (Sections 7.1 and 7.2) use synthetic returns drawn from a spiked covariance model calibrated to the S&P 500 regime ($n/T \approx 0.4$, factor structure matching the Marchenko–Pastur prediction; seed fixed at 42). The condition-number and k-sensitivity results (Sections 3 and 6) use the actual S&P 500 return series ($n = 494$, $T = 1213$); the out-of-sample backtests (Section 7.5) additionally use the FTSE 100 series.

7.1 Solver comparison

Four solvers are applied to the same problem: minimise $w^\top T_0^{\mathrm{RMT}} w - \rho\,\hat{\mu}^\top w$ subject to $w \geq 0$, $\mathbf{1}^\top w = 1$ over a 50-point efficient frontier. Synthetic equity data: $n = 500$ assets, $T = 1250$ days, $k = 22$ signal factors. Active-set sizes averaged $n_a = 42$ across the frontier (warm start).

- **CVXPY** (Clarabel backend): general-purpose modelling-language reference, cold start only.
- **Clarabel (direct)**: the same interior-point solver called through its native API with pre-assembled sparse problem data, bypassing CVXPY's construction overhead; cold start only.
- **KKT-Cholesky**: the same primal-dual outer loop as Algorithm 1, but with the inner solve replaced by explicitly assembling $T_{0,\mathcal{A}}^{\mathrm{RMT}}$ as a dense $n_a \times n_a$ matrix via a rank-k symmetric update and then factoring by Cholesky.
- **Woodbury**: Algorithm 1 as stated – the inner solve uses the Woodbury identity and never assembles an $n_a \times n_a$ matrix.

	Cold start		Warm start	
Solver	Total (s)	ms/pt	Total (s)	ms/pt
Synthetic, $n = 500$, $T = 1250$, $k = 22$, 50-point sweep (total / ms per point)				
CVXPY (Clarabel)	88.5	1769.7	–	
KKT-Cholesky ($k = 22$)	0.3233	6.5	0.0403	0.8
Woodbury ($k = 22$)	0.0168	0.3	0.0082	0.2
S&P 500, $n = 494$, $T = 1213$, $k = 23$, single min-var solve (seconds / ms)				
CVXPY (Clarabel)	1.573	1573.2	–	
KKT-Cholesky ($k = 23$)	0.0066	6.6	0.0004	0.4
Woodbury ($k = 23$)	0.0002	0.2	0.0001	0.1

Table 3: Solver timings on two datasets. *Top panel*: synthetic equity data, 50-point frontier sweep; totals and ms per frontier point. *Bottom panel*: real S&P 500 data ($n = 494$, $T = 1213$), single minimum-variance solve; the 'ms/pt' column gives ms per solve. Woodbury and KKT-Cholesky produce identical portfolios to machine precision; agreement with the interior-point solutions is reported in the text.

Correctness Woodbury and KKT-Cholesky execute the same outer loop on the same subproblems and return identical portfolios (maximum frontier-volatility difference 9×10^{-15} percentage points). Against the independent interior-point solution the agreement is $\|w_{\mathrm{WB}} - w_{\mathrm{Clarabel}}\|_\infty = 2.5 \times 10^{-4}$ at the minimum-variance point and 3.6×10^{-5} at mid-frontier (1.4×10^{-4} on the S&P 500 solve).

The discrepancy is attributable to interior-point truncation, not to the active-set method: the Woodbury solution attains a strictly *lower* objective value than the Clarabel solution (relative difference 3.3×10^{-5}), so the direct solution is the better optimum of the two.

Timings On synthetic data (top panel), Woodbury is 19× faster than KKT-Cholesky on a cold start and 5× faster warm. Both methods execute the same outer active-set loop; the difference is entirely in the inner solve. KKT-Cholesky assembles a 42×42 matrix at each outer step via a rank-k symmetric update and factors by Cholesky. Woodbury instead operates on a $k \times k = 22 \times 22$ correction matrix W (Remark 4.3). The cold-start ratio exceeds the warm-start ratio because the cold solver takes more outer active-set steps before converging (~ 12 cold vs ~ 2 warm) at systematically larger active-set sizes, where the per-step gap between $O(n_a^2 k + n_a^3)$ and $O(n_a k^2)$ is widest.

The bottom panel confirms the result on real S&P 500 data ($n = 494$, $T = 1213$, $k = 23$, $n_a = 63$ active assets at the minimum-variance solution): Woodbury is more than 20× faster than KKT-Cholesky on a single cold minimum-variance solve (28× on unrounded timings) and 5× faster warm. For the single-solve setting the warm start uses the converged active set from an immediately preceding cold solve at the same $\rho = 0$, measuring the incremental re-solve cost when the covariance is refreshed but the optimal active set has not changed – the scenario for daily rebalancing. The warm-start advantage is not an artefact of the synthetic data-generating process: the Markowitz frontier is piecewise linear in weight space with generically single-asset exchanges at breakpoints (Section 4), for any return distribution.

Interior-point baselines The fair interior-point reference is the direct Clarabel call: just over 60 ms per solve, more than two orders of magnitude above Woodbury cold – attributable to the $n \times n$ dense quadratic form the interior-point method must factor at each of its iterations, against the $k \times k$ correction matrix of the direct method. The CVXPY route adds a further $\sim 25\times$ on top (1.6 s per solve) for Python-level problem construction and matrix assembly into the solver's internal format. We report both: the CVXPY number is what a practitioner observes out of the box, the direct-Clarabel number is what the interior-point algorithm itself costs; neither is a measure of Clarabel's quality on general problems.

7.2 Scaling with n

Figure 1 shows wall-clock time vs. n ($T = 1250$ fixed) for four configurations: KKT-Cholesky (no preprocessing), Woodbury solve only, Woodbury with dense preprocessing, and Woodbury with randomised SVD preprocessing.

Two caveats frame this experiment. First, the growth law $k \approx n/21$ is observed on the real data at a single size ($k = 23$ at $n = 494$) and is *built into* the synthetic generator; the large-n conclusions are conditional on it. Second, under that law both methods scale cubically – Cholesky as $O(n_a^3)$ with $n_a \propto n$, Woodbury as $O(n_a k^2)$ with $k \propto n$ – so the asymptotic gap is a constant factor, not an order. The *measured* ratio nevertheless grows from 26× at $n = 500$ to 86× at $n = 3000$, consistent with the $n_a \times n_a$ working set falling out of cache while the $k \times k$ correction matrix does not. If k is fixed externally as n grows (Remark 3.3), Woodbury gains a full asymptotic order over Cholesky.

Dense preprocessing dominates at large n. At $n = 3000$ the full eigendecomposition takes 1.25 s, versus 0.22 s for KKT-Cholesky and 0.003 s for the Woodbury solve. Dense preprocessing is impractical for $n > 750$.

Randomised SVD preprocessing is competitive throughout. The rSVD path (0.051 s at $n = 3000$) is 25× faster than dense preprocessing; combined with the Woodbury solve (0.003 s), the total

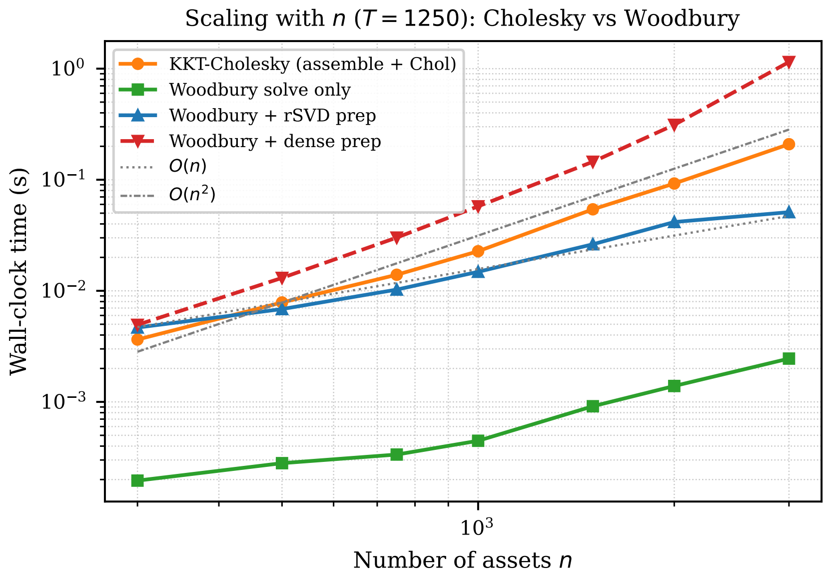

Figure 1: Wall-clock time vs. n (log-log, $T = 1250$). KKT-Cholesky grows faster than Woodbury as n increases because n_a grows with n and the $O(n_a^3)$ Cholesky term dominates. Dashed reference lines $O(n)$ and $O(n^2)$ anchored at $n = 500$.

pipeline is 4.2× faster than KKT-Cholesky alone (0.224 s) – confirming the headline claim of the abstract. For $S \geq 2$ portfolio solves (any frontier sweep, monthly rebalance sequence, etc.) the preprocessing cost amortises further, making the pipeline advantage grow with the number of solves.

7.3 Efficient frontier

Figure 2 shows the 50-point efficient frontier on synthetic equity data ($n = 500$, $T = 1250$), coloured by active-asset count. The minimum-variance point (star) has $n_a = 31$ active assets; the high-return end has $n_a = 97$. The rapid decrease in active assets away from the mean-variance efficient region is characteristic of the long-only constraint and explains the warm-start efficiency: adjacent frontier points share all but a few active assets.

7.4 Exact frontier via the critical line algorithm

Remark 4.1 observed that on each active set the solution is affine in ρ, so the exact frontier can be traced breakpoint-to-breakpoint with one Woodbury solve per segment. We implement this sweep and run it on the Section 7.1 problem ($n = 500$, $k = 22$, $\rho \in [0, 2]$). The exact frontier has 144 breakpoints and is traced in 9 ms – the same cost as the 50-point warm grid sweep (9 ms, Table 3) while returning the complete piecewise-affine solution path rather than 50 samples of it. Evaluating the CLA path at the 50 grid values reproduces the grid-swept portfolios up to the grid solver's KKT tolerance (maximum weight difference 2.7×10^{-3}); at every grid point the CLA portfolio attains an objective value at least as low as the grid solver's, confirming that the residual difference is grid-solver truncation at near-degenerate breakpoints, not an error in the path. For a fixed covariance the exact sweep is therefore strictly preferable; the grid sweep remains the tool of choice when consecutive solves differ in the data rather than in ρ (Remark 4.1).

n	k	Cholesky (s)	WB solve (s)	WB+dense (s)	WB+rSVD (s)
300	11	0.0033	0.0002	0.0049	0.0046
500	22	0.0079	0.0003	0.0134	0.0070
750	34	0.0155	0.0004	0.0315	0.0105
1000	47	0.0243	0.0005	0.0571	0.0158
1500	72	0.0572	0.0010	0.1463	0.0276
2000	90	0.0964	0.0014	0.3242	0.0443
3000	123	0.2240	0.0026	1.2490	0.0532

Table 4: Scaling data ($T = 1250$ fixed). k is the MP-detected signal factor count. At $n = 3000$ Woodbury is 86× faster than KKT-Cholesky (solve only) and the total pipeline including rSVD preprocessing (0.053 s) is 4.2× faster than KKT-Cholesky (0.224 s). The comparison is conservative: KKT-Cholesky also requires the eigenpairs to assemble T_0^{RMT}, but is charged no preprocessing here.

7.5 Out-of-sample performance

The speed results above are only useful if the RMT-cleaned estimator is statistically sound on real data. We therefore run rolling-window long-only minimum-variance backtests on two universes: the S&P 500 ($n/L \approx 0.98$, deliberately high-dimensional) and the FTSE 100 ($n/L \approx 0.17$, a comfortably sampled regime in which cleaning should matter less). In both cases: estimation window $L = 504$ days, monthly rebalancing, buy-and-hold drift between rebalances. Assets with zero full-sample variance or more than 20% exact-zero returns (delisted names forward-filled flat, late listings padded with zeros) are excluded – a zero-variance asset absorbs the entire minimum-variance portfolio and renders the comparison meaningless. Four strategies are compared: the RMT-CRE portfolio of this paper ($\alpha = 1$, solved by Algorithm 1); linear shrinkage to the scaled identity at the practitioner level $\alpha = 0.5$ and at the Ledoit–Wolf oracle intensity [10]; and equal weight.

Strategy	OOS vol (% ann.)	Sharpe	Turnover (%)
RMT ($\alpha = 1$)	10.77	1.193	14.5
LW-0.5 ($\alpha = 0.5$)	10.52	1.219	13.3
LW-oracle	10.50	1.538	21.7
Equal-weight	15.67	1.086	0.0

Table 5: Out-of-sample rolling-window backtests (estimation window $L = 504$ days, monthly rebalancing). Turnover is the mean one-way turnover per rebalance. The per-window MP-detected k is shown in each panel header.

The pattern is consistent across both universes. All three minimum-variance strategies reduce realised volatility by roughly a third relative to equal weight. On the S&P 500 the RMT-CRE portfolio (9.83%) sits within 0.1 percentage points of the linear-shrinkage alternatives (9.75% and 9.74%) with comparable turnover; on the FTSE 100 it is the lowest-volatility strategy outright (10.88% against 10.97% and 11.06%), though the margin is again within noise. In the well-sampled FTSE regime the three estimators nearly coincide, as expected when n/L is small. We do not claim statistical superiority for CRE – only that the estimator whose structure enables the direct solver carries no measurable out-of-sample penalty in either regime, despite using no tuning parameter beyond the MP threshold. This is consistent with the larger comparative literature on RMT cleaning [3, 14].

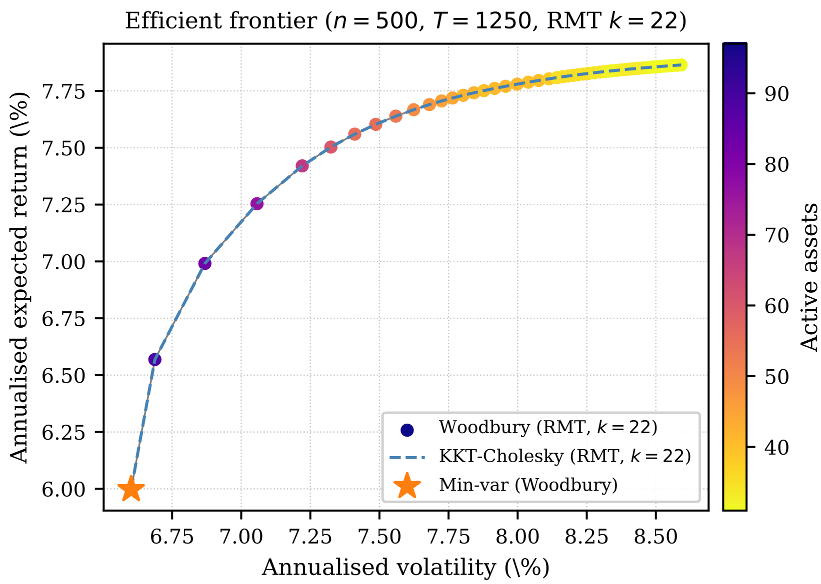

Figure 2: Efficient frontier ($n = 500$, $T = 1250$, $k = 22$, $\alpha = 1$). Scatter: Woodbury warm, coloured by active-asset count (star = minimum-variance portfolio, $n_a = 31$). Dashed: KKT-Cholesky warm (same portfolio to numerical precision, plotted to confirm equivalence). Both sweeps take under 0.05 s warm; see Table 3 for timings.

Reproducibility All experiments are produced by a single script (`experiment_rmt.py`) in the repository of the accompanying open-source package; random seeds are fixed (42 for the synthetic generator, 0 for the randomised SVD), and the S&P 500 and FTSE 100 return panels are included in the repository.

8 Conclusions

The sample covariance of equity returns mixes a small number of well-estimated signal eigenvalues with several hundred poorly estimated noise eigenvalues. Replacing the noise eigenvalues with their average – the trace-preserving CRE cleaning prescribed by the Marchenko–Pastur analysis [9, 3] – reduces the condition number by a factor of roughly 300 on S&P 500 data, and its low-rank-plus-identity form admits an exact direct solve via the Woodbury identity.

This paper makes three algorithmic contributions and one empirical contribution.

Direct Woodbury portfolio solver. The primal-dual active-set loop requires an inner solve of $T^{\mathrm{RMT}}_{0,\mathcal{A}} w = b$ at each outer step. The naive approach assembles the $n_a \times n_a$ active-set system matrix and solves by Cholesky at cost $O(n_a^2 k + n_a^3)$ per step. The Woodbury identity reduces this to $O(n_a k^2 + k^3)$ by working with the $k \times k$ correction matrix W instead – cheaper in flops precisely when $k < n_a$ (Proposition 4.2), which holds throughout the long-only equity regime. The measured wall-clock advantage is 5× warm and 19× cold vs KKT-Cholesky on a 50-point synthetic frontier sweep, and more than 20× cold on a real S&P 500 minimum-variance solve. Against interior-point solvers the advantage is two orders of magnitude (Clarabel called natively) and over three orders of magnitude (CVXPY out of the box). The returned weights agree with interior-point solutions to solver accuracy and attain marginally lower objective values. The same Woodbury kernel powers an exact critical-line sweep that traces all 144 frontier breakpoints for the cost of a 50-point grid

(Section 7.4).

Randomised SVD preprocessing. Computing the top-k eigenpairs of $\hat{\Sigma}$ via a dense eigendecomposition costs $O(n^2T + n^3)$ and requires $O(n^2)$ storage. A randomised truncated SVD applied directly to the return matrix X computes the same eigenpairs in $O(nTk)$ time and $O(nk)$ storage; the measured portfolio impact of the approximation is below 3 bp in any weight, an order of magnitude below the $k \pm 1$ threshold uncertainty (Section 6). At $n = 3000$, randomised preprocessing is $25\times$ faster than dense, and the full pipeline (rSVD + Woodbury solve) is $4\times$ faster than the KKT-Cholesky solve alone – with no preprocessing charged to the latter. For daily rolling windows an incremental rank-two update of the tracked eigenpairs replaces the re-sketch at a $23\times$ lower per-day cost, with median portfolio drift of 7 bp over 100 days (Section 5).

Scope of the structural trick. Ridge terms, diagonal loadings, externally specified factor models, and linear equality constraints preserve the Woodbury structure (Remark 4.2); nonlinear shrinkage and general rotationally invariant estimators [12, 13, 3] do not, because they modify the full spectrum. CRE clipping is the member of the RMT cleaning family that trades a known statistical refinement for an exploitable algebraic structure.

Empirical validation. The MP threshold is an insensitive hyperparameter: misjudging k by one factor moves portfolio weights by tens of basis points and volatility by less than one basis point, and the three-point audit costs milliseconds. Rolling out-of-sample backtests on two universes – the high-dimensional S&P 500 ($n/L \approx 0.98$) and the well-sampled FTSE 100 ($n/L \approx 0.17$) – show the CRE portfolio within 0.1 volatility points of linear shrinkage at the practitioner and oracle intensities (and lowest outright on the FTSE), with all minimum-variance strategies roughly a third less volatile than equal weight: the estimator that enables the fast solver carries no measurable out-of-sample penalty in either regime.

Practical recommendations For $n \leq 500$, either preprocessing path is practical; use dense if simplicity matters. For $n > 1000$, use randomised SVD. For any sweep of $S \geq 2$ solves (efficient frontier, rebalance sequence, sensitivity audit), the preprocessing cost amortises to $O(nTk/S)$ per solve. The Woodbury solver is preferred over KKT-Cholesky whenever $k < n_a$, which holds throughout the parameter regime of long-only institutional equity portfolios.

References

[1] Jinho Baik, Gérard Ben Arous, and Sandrine Péché. Phase transition of the largest eigenvalue for nonnull complex sample covariance matrices. *Annals of Probability*, 33(5):1643–1697, 2005.

[2] Jean-Philippe Bouchaud and Marc Potters. *Theory of Financial Risk and Derivative Pricing: From Statistical Physics to Risk Management.* Cambridge University Press, 2nd edition, 2003.

[3] Joël Bun, Jean-Philippe Bouchaud, and Marc Potters. Cleaning large correlation matrices: tools from random matrix theory. *Reviews of Modern Physics*, 89(3):035005, 2017.

[4] Noureddine El Karoui. Spectrum estimation for large dimensional covariance matrices using random matrix theory. *Annals of Statistics*, 36(6):2757–2790, 2008.

[5] Gene H. Golub and Charles F. Van Loan. *Matrix Computations.* Johns Hopkins University Press, 4th edition, 2013.

[6] Nathan Halko, Per-Gunnar Martinsson, and Joel A. Tropp. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review*, 53(2):217–288, 2011.

[7] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, 2nd edition, 2013.

[8] Iain M. Johnstone. On the distribution of the largest eigenvalue in principal components analysis. *The Annals of Statistics*, 29(2):295–327, 2001.

[9] Laurent Laloux, Pierre Cizeau, Jean-Philippe Bouchaud, and Marc Potters. Noise dressing of financial correlation matrices. *Physical Review Letters*, 83(7):1467–1470, 1999.

[10] O. Ledoit and M. Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.

[11] O. Ledoit and M. Wolf. Analytical nonlinear shrinkage of large-dimensional covariance matrices. *Annals of Statistics*, 48(5):3043–3065, 2020.

[12] Olivier Ledoit and Sandrine Péché. Eigenvectors of some large sample covariance matrix ensembles. *Probability Theory and Related Fields*, 151(1–2):233–264, 2011.

[13] Olivier Ledoit and Michael Wolf. Nonlinear shrinkage estimation of large-dimensional covariance matrices. *The Annals of Statistics*, 40(2):1024–1060, 2012.

[14] Marcos López de Prado. *Advances in Financial Machine Learning*. Wiley, Hoboken, NJ, 2018.

[15] Vladimir A. Marchenko and Leonid A. Pastur. Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4):457–483, 1967.

[16] H. Markowitz. The optimization of a quadratic function subject to linear constraints. *Naval Research Logistics Quarterly*, 3(1–2):111–133, 1956.

[17] Andras Niedermayer and Daniel Niedermayer. Applying Markowitz's critical line algorithm. In John B. Guerard, editor, *Handbook of Portfolio Construction: Contemporary Applications of Markowitz Techniques*, pages 383–400. Springer, Boston, MA, 2010.

[18] Jorge Nocedal and Stephen J. Wright. *Numerical Optimization*. Springer, New York, 2nd edition, 2006.

[19] André F. Perold. Large-scale portfolio optimization. *Management Science*, 30(10):1143–1160, 1984.

[20] Vasiliki Plerou, Parameswaran Gopikrishnan, Bernd Rosenow, Luis A. Nunes Amaral, and H. Eugene Stanley. Universal and nonuniversal properties of cross correlations in financial time series. *Physical Review Letters*, 83(7):1471–1474, 1999.

[21] T. Schmelzer. fast-minimum-variance: Solving minimum variance portfolios fast. `https://github.com/Jebel-Quant/fast_minimum_variance`, 2026. MIT License.

[22] Thomas Schmelzer. cvxcla: Critical line algorithm for portfolio optimisation. `https://github.com/cvxgrp/cvxcla`, 2024. MIT License.